

State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

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11

retrieval branches

146

ranker features

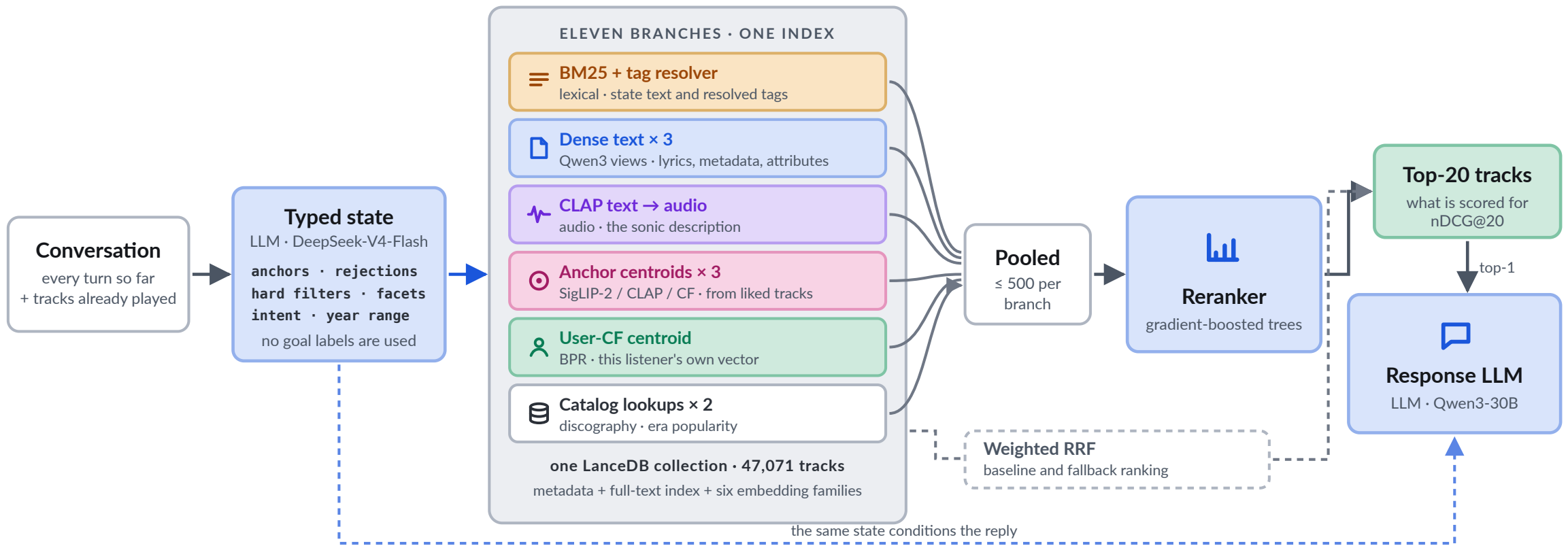
0.3811

Blind-B composite

29_{/40}

where that finished

One state, eleven branches, one learned ranker



What the ranker leans on

FEATURE FAMILY	#	GAIN
Bi-encoder cosine	1	59.0%
Per-branch rank and score	66	11.5%
Session and artist affinity	18	10.7%
Popularity · state · tags	31	9.2%
Other similarity cosines	17	8.9%
Constraints and rejections	13	0.7%

SETUP

LightGBM LambdaMART, a gradient-boosted tree ranker. LR 0.025, 127 leaves, truncation 200, 5-fold user-grouped CV. ~20.6M candidate rows.

READ IT CAREFULLY

Split gain describes the fitted model, not recommendation quality. But **constraints at 0.7%** is real: the ranker leans on similarity, not on what the listener asked for.

GUARDS AFTER SCORING

Exact-track pin when a specific track is named; a final artist-constraint check.

How it scored

OFFICIAL CODABENCH SPLIT	NDCG@20	CAT.	LEX.	JUDGE	COMPOSITE
Blind-A • development phase	0.4380	0.0313	0.8028	4.70	0.5799
Blind-B • final, 29th of 40	0.2537	0.0315	0.7862	3.30	0.3811

NDCG@20

-42%

0.4380 → 0.2537

JUDGE (1-5)

-1.40

4.70 → 3.30

DIVERSITY TERMS

flat

category and lexical barely moved

Blind-B also withholds the conversation-goal annotations — but the deployed ranker never used them at inference, so that is not the explanation.

What is not ideal

We tuned on what we measured.

The 8,000-turn development set was both our training and our evaluation data, and the response prompt was picked against the Blind-A leaderboard: 4.70 there, 3.30 on Blind-B.

We extracted more than we used.

The state we read off a conversation was far richer than the state that reached retrieval or ranking. Most fields never became a filter or a ranking signal at all.

Nothing from outside the catalog.

No outside source for what a track or artist is known for, and no tempo, key or soundtrack. Past one CF vector, no listening behaviour — no co-occurrence, no transitions.

The reply was never checked.

Asked for a track we do not have, the generator described a different song by the same artist instead of saying so.

Asked for, served

the Kamelot track
from **Silverthorn**



Kamelot – Fallen Star, from **Haven**
CONSTRAINT NEVER APPLIED

exactly **126.70 bpm**,
in G minor



no tempo or key field exists to match on
NOTHING IN THE CATALOG

“Watercolors”
by Pat Metheny



Metheny – Alfie, described as if it were
RESPONSE LLM

Silverthorn is in the catalog. The third is a response failure, not a retrieval one – the track simply does not exist in the 47,071.

The target often ignores the request

User: "...something with a similar chill, electronic vibe, but **from a different artist?**" (just played: Bonobo)

Target: **Bonobo – "Jets"** · annotated MOVES toward goal

Next turn: "'Jets' is cool, but I was actually hoping for something **from a different artist this time...**"

Target: **Bonobo – "Pieces"** · annotated MOVES toward goal

Ground truth is simply **whatever played next**, in a session an LLM wrote. Our own model anchors on the just-played artist — we think it learns that here.

What we would do differently

- 1 **Hold out a clean split before touching anything.** Free, and it would have changed every selection decision we made.
- 2 **Make the state executable.** Rejections and constraints carried 0.7% of model gain — most of what the extractor understood never became a filter.
- 3 **Import knowledge the catalog lacks** — tempo, key, soundtrack and film/TV links — so requests naming them have something to match.
- 4 **Check the response against the state** before emitting it: flag unavailable tracks and unmet constraints instead of overselling a substitute.
- 5 **Decide what to do when the benchmark and your judgment disagree.** We still do not have a principled answer.

Everything is released

IN THE REPO

- Pipeline code, configs, trained models
- Cached per-turn extraction states
- The 113,393 re-judged labels
- A frozen-LLM-cache replay bundle

ZERO-CREDENTIAL REPLAY

Every LLM call is served from a curated frozen cache that **hard-fails on a miss** rather than falling through to a paid call. Both submissions reproduce without any API key.



Code, models, data

[github.com/npatta01/
music-crs-2026](https://github.com/npatta01/music-crs-2026)

